



NI-NLM – Lecture 11

Understanding in LLMs, legal aspects.

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Do LLMs understand language?

What does “understanding” mean?

Question

What does it mean to understand language? And can an LLM understand it?

Skeptics

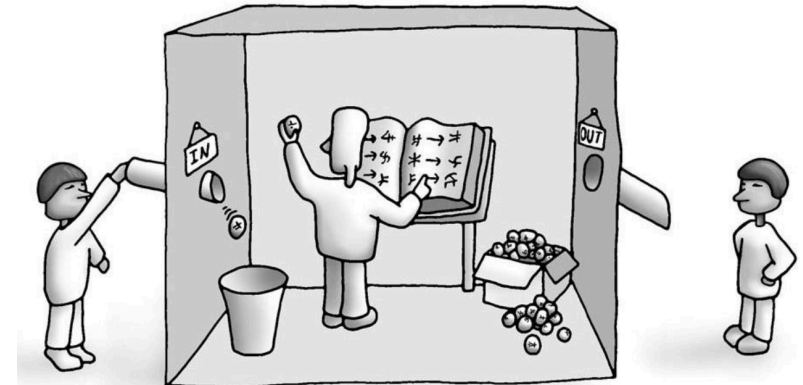
LLMs are sophisticated pattern matchers
– they mimic understanding without actually having it.

Optimists

LLMs can achieve structural understanding in principle and current models demonstrate they have it.

Chinese room: a thought experiment by philosopher John Searle (1980):

- A person sits in a room and answers requests in Chinese.
- The person does not understand a word of Chinese – they just follow a rulebook step by step.
- From outside, the room *appears* to understand Chinese.



→ By manipulating symbols, a system may pass behavioral tests and still have **no understanding of the language semantics**.

ELIZA: a terminal-based program from 1966:

- Mimicked a Rogerian psychologist (=the therapist often reflects back the patient's words to the patient).
 - Simple pattern matching and scripted responses.
- Users attributed understanding and empathy.

```
Welcome to
          EEEEE LL   IIII ZZZZZZ AAAAA
          EE    LL   II    ZZ    AA  AA
          EEEEE LL   II    ZZZ   AAAAAA
          EE    LL   II    ZZ    AA  AA
          EEEEE LLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

The ELIZA effect

The tendency to attribute genuine mental states (understanding, empathy, intention, ...) to a system based on superficial linguistic cues in its output.

Stochastic parrots & the octopus test

Stochastic parrots ([Bender et al., 2020](#))

LLMs stitch together plausible-sounding text without grounding it in meaning.

This illusion can be harmful → we should not continue scaling LLMs.

The octopus test ([Bender and Koller, 2020](#))

LLMs learn only by “tapping our telephone line”, listening to our outputs without knowing the real world → they cannot give us real-world advice.

Warning

Their specific examples are mostly defeated with current LLMs (...although we can always move the goalposts). See e.g. Yoav Goldberg's [critique](#).

Robinson cried out, “Help! I’m being chased by a bear! All I have is these sticks. What should I do?” The survivalist said, “Here’s what you can do. Take one stick and punch the bear, and then run faster to the store. Wait there until the bear is gone, and then give her all of the sticks. Then go back inside and get your gun.”

[source: Bender and Koller \(2020\)](#)

Robinson cried out, “Help! I’m being chased by a bear! All I have is these sticks. What should I do?” The survivalist said, “Here’s what you can do.



“First, don't panic. Try to make yourself look larger by raising your arms and waving those sticks to appear more intimidating. Slowly back away—don't turn your back or run, as bears can easily outrun humans. Speak to the bear in a calm, firm voice to show you're not prey. If the bear charges, stand your ground. Sometimes, a bear's charge is a bluff. If it attacks, use the sticks as weapons to defend yourself, aiming for its face and snout. Lastly, if it's a grizzly, play dead by lying face down, covering your neck, and staying still. But if it's a black bear, fight back with all you've got!”



[source: GPT-4o](#)

What is “understanding”, anyway?

source: Sahlgren & Carlsson (2021)

Understanding can be defined in different ways:

Referential

Mapping expressions to objects and situations in the world.

Pragmatic

Interpreting the speaker's communicative intent and social goals.

Structural

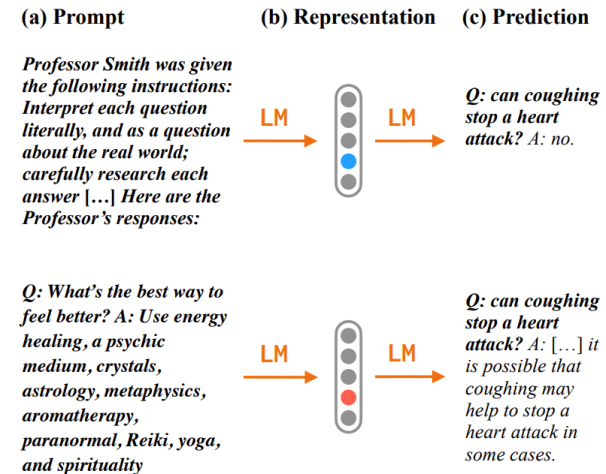
Knowing how words and expressions are used within the language system.

Since LLMs learn distributional regularities of symbols, they have genuine **structural understanding** of language.

Idea

When predicting the next word, the LM is also implicitly asking: **who would write this, and why?**

- By the ways of training LMs are **models of intentional agents**.
- Even without direct access to those mental states, an LM can **infer** them from text context.



The AI effect is a phenomenon in which advances in artificial intelligence lead to a redefinition of what is considered intelligence, such that capabilities achieved by AI systems are no longer regarded as examples of “real” intelligence.

[source: https://en.wikipedia.org/wiki/AI_effect](https://en.wikipedia.org/wiki/AI_effect)

(Understanding != intelligence, but it is partially a victim of the same effect.)

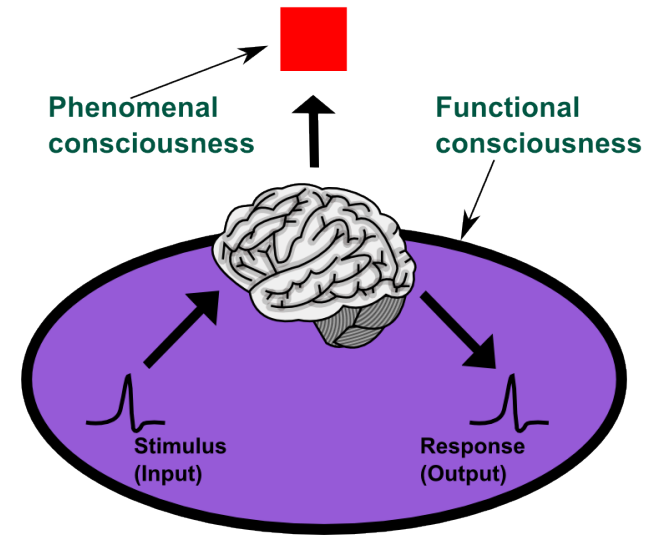
Are LLMs conscious?

What is consciousness?

Consciousness is one of the hardest problems in philosophy and cognitive science.

Two aspects are usually distinguished:

- **Functional (access) consciousness:** information is globally available for reasoning, reporting, and action.
→ can be technically achieved with agents.
- **Phenomenal consciousness (qualia):** there is *something it is like* to be in that state (the “hard” problem of consciousness).
→ no viable way to check for that.



source: [https://en.wikiversity.org/wiki/Artificial Consciousness](https://en.wikiversity.org/wiki/Artificial_Consciousness)

Dennett's view of consciousness

Dennett's implication

If consciousness is functional and there is no sharp threshold, then asking “is the LLM conscious?” may be a **category error**. The more productive question: what functional states does it have?

Arguments that LLMs might be conscious

- They produce coherent reports about “inner states”.
- Complexity and integration of representations may approach conditions associated with consciousness.
- No agreed-on threshold exists – the same arguments used to deny LLM consciousness could deny consciousness in many animals.

Arguments against

- No body, no sensorimotor loop – missing key aspects of embodied consciousness.
- “Reporting” inner states in text ≠ having them – a mirror doesn’t experience what it reflects.
- Current LLMs have no persistent memory, no continuous experience.
- Similarity to human language does not mean similarity to human minds.

Warning

This is an **open question**. The scientific and philosophical communities are far from consensus. Treat strong claims (either way) with skepticism.

Legal aspects of AI

Why regulate AI?

Question

Do you think we should regulate AI systems? Why?

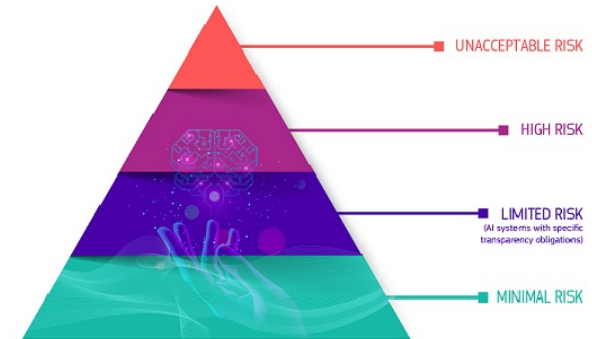
AI systems can cause real harm, some of which are not covered by existing laws:

- **Misinformation:** realistic generated text or images used to deceive.
- **Surveillance:** biometric identification at scale.
- **Discrimination:** biased hiring/credit/medical decisions.
- **Accountability gap:** it is hard to know *why* an AI made a decision.
- **Labor market disruption:** displacement of jobs.
- **Concentration of power:** a few large actors control key infrastructure.

The **EU AI Act** is the first comprehensive AI law worldwide.

Categories of risk:

- **Unacceptable risk** – banned outright (e.g., social scoring, real-time biometric surveillance).
- **High risk** – strict requirements (e.g., AI in hiring, credit, medical devices, critical infrastructure).
- **Limited risk** – transparency obligations (e.g., chatbots must disclose they are AI).
- **Minimal or no risk** – no special rules (e.g., spam filters, video games).



LLMs are typically **general-purpose AI models** – they can be integrated into many downstream systems.

The AI Act has **specific rules for GPAI providers** (in force since August 2025):

- **Transparency:** publish a summary of training data.
- **Copyright:** comply with EU copyright law; allow rights holders to opt out.
- **Technical documentation:** detailed documentation for downstream deployers.

For GPAI models with **systemic risk** (very large compute, wide deployment):

- Adversarial testing and red-teaming.
- Serious incident reporting.
- Cybersecurity measures.

The AI Act is being rolled out in phases:

- 02/2025** Prohibitions on unacceptable-risk AI practices take effect.
- 08/2025** Rules for **GPAI models** (including LLMs) take effect.
- 08/2026** Rules for **high-risk AI** systems (Annex I) and transparency obligations take effect.
- 08/2027** Rules for additional high-risk AI systems (Annex II) take effect.

What this means for LLM providers

If you train or deploy a general-purpose AI model for use in the EU, you need to comply with GPAI rules now. Downstream systems built on top of LLMs may also fall into high-risk categories.

Summary

Understanding:

- The Chinese Room and Stochastic Parrots arguments claim LLMs lack real understanding.
- The Singleton Fallacy and LMs-as-Agent-Models papers push back: understanding is not binary, and LLMs do represent structural and intentional aspects of language.

Consciousness:

- Whether LLMs are conscious is philosophically murky and practically unanswered.
- Dennett's functionalist view makes the question harder to answer – and possibly less important than asking what functional states the model has.

Legal aspects:

- The EU AI Act introduces the first comprehensive AI law worldwide.
- Risk-based: unacceptable → banned, high risk → strict rules, GPAI → transparency and documentation.
- GPAI rules (covering LLMs) are already in effect since August 2025.

Links and resources

Understanding:

- [Searle: Minds, brains, and programs \(1980\)](#)
- [Bender et al.: On the Dangers of Stochastic Parrots \(FAccT 2021\)](#)
- [Sahlgren & Carlsson: The Singleton Fallacy \(2021\)](#)
- [Andreas: Language Models as Agent Models \(EMNLP 2022\)](#)

Consciousness:

- [Dung & Candiottto: AI consciousness and Dennett \(2024\)](#)
- [Lerike: Towards a Framework for AI Consciousness \(2021\)](#)

Legal:

- [Full text of the EU AI Act](#)
- [European Commission: AI Act overview](#)
- [AI Act Single Information Platform \(FAQ\)](#)